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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

Java Programming Project Presentation

- Project Title: Personal Contact Directory with Search and Update Features
- COURSE : JAVA PROGRAMMING LAB / MINI PROJECT
- Student Name & Roll No: Roshini Kumari P [25bcs142]
- Department: Computer Science
- Mentor / Guide: Ashwanandini

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Introduction

Thisprojectis a Java-basedapplication to manage personal contacts efficiently.
It allows users to store, search, update, and delete contact details.
Helps overcome the limitations of manual contact storage.
Useful for organizing phone numbers, emails, and addresses in one place.

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Objective

Maingoals of theproject:
To develop a contact management system using Java
To implement search and update functionalities
To apply object-oriented programming concepts
To improve problem-solving and programming skills

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Problem Statement

Managing contacts manually istime-consuming and inefficient
Difficult to search or update specific contact details quickly
Risk of losing data without proper storage
This project provides a digital solution for easy contact management

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System Requirements

Hardware Requirements:
Processor: Intel i3 or higher
RAM: 4 GB minimum
Storage: 500 MB free space
Software Requirements:
OS: Windows / Linux
Programming Language: Java
IDE: Eclipse / IntelliJ / NetBeans

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System Design

The systemconsistsofmultiple modules:
User Interface
Contact Management
Search & Update Module
File Storage
Include:
Flowchart
Block Diagram
Algorithm

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Module Description

InputModule:
Accepts userdetails(name, phonenumber, email)
Processing Module:
Handles adding, searching, updating, deleting contacts
Output Module:
Displays contact details to the user
File Handling Module:
Stores and retrieves contact data from files

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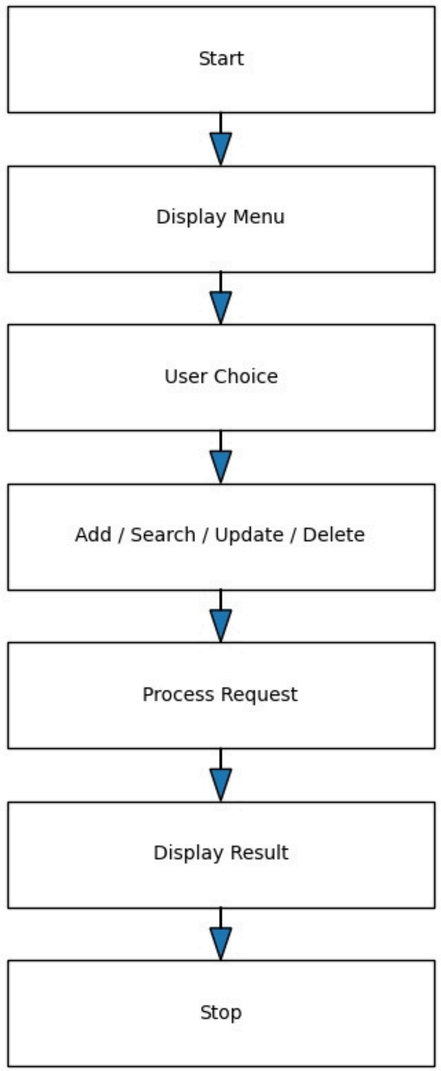
Algorithm / Flowchart

Start Display menu options Take user choice If add → enter contact details If search → find contact If update → modify contact If delete → remove contact Display result Stop


Start
Display Menu
User Choice
Add / Search / Update / Delete
Process Request
Display Result
Stop

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Implementation

Techniques used:
Classes and Objects (OOP concepts)
Arrays / ArrayList for storing contacts
Loops and conditional statements
File handling (FileWriter, FileReader, BufferedReader)

```
import java.util.*;

class Contact {
String name, phone;

Contact(String n, String p) {
name = n;
phone = p;
}
}

public classContactDirectory {
public staticvoidmain(String[] args) {
ArrayList<Contact>list=new ArrayList<>();
Scanner sc = new Scanner(System.in);

// Add Contact
System.out.print("Enter Name: ");
Stringname=sc.nextLine();
System.out.print("Enter Phone: ");
Stringphone=sc.nextLine();
list.add(newContact(name, phone));

// Search Contact
System.out.print("Search Name: ");
String search = sc.nextLine();
for(Contactc:list) {
if(c.name.equalsIgnoreCase(search)) {
System.out.println(c.name + " - " + c.phone);
}
}
}
}
```

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Output Screens

Menu display screen
Add contact screen
Search result screen
Update/Delete confirmation

MENU
1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Exit

```
ADD CONTACT
Enter Name: John
Enter Phone: 9876543210
Contact Added Successfully!
```

SEARCH RESULT
Search Name: John
Found: John - 9876543210

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Results & Discussion

Successfullystored and managedcontacts

Fastsearch and updateoperations

Accurate results obtained

Challenges faced:

Handling file operations

Managing user inputs

Solutions:

Used exception handling

Improved logic design

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Conclusion & Future Enhancement

Conclusion:

The project meetsall objectives

Demonstrates Java programming and OOP concepts

Provides an efficient contact management system

Future Enhancements:

Add GUI using Java Swing or JavaFX

Use database (MySQL) for storage

Add login/security features

Cloud-based access

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References / Acknowledgement

References:

Java Programming – E. Balagurusamy

Oracle Java Documentation

Online tutorials

Acknowledgement:

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Thank You